



Data & metadata tests

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Metadata test

<https://github.com/PetrColinSky/AdriaArray>

- list of AdriaArray stations: Petr's tables + _ADARRAY
- metadata downloaded from EIDA on Mar 04, 2025

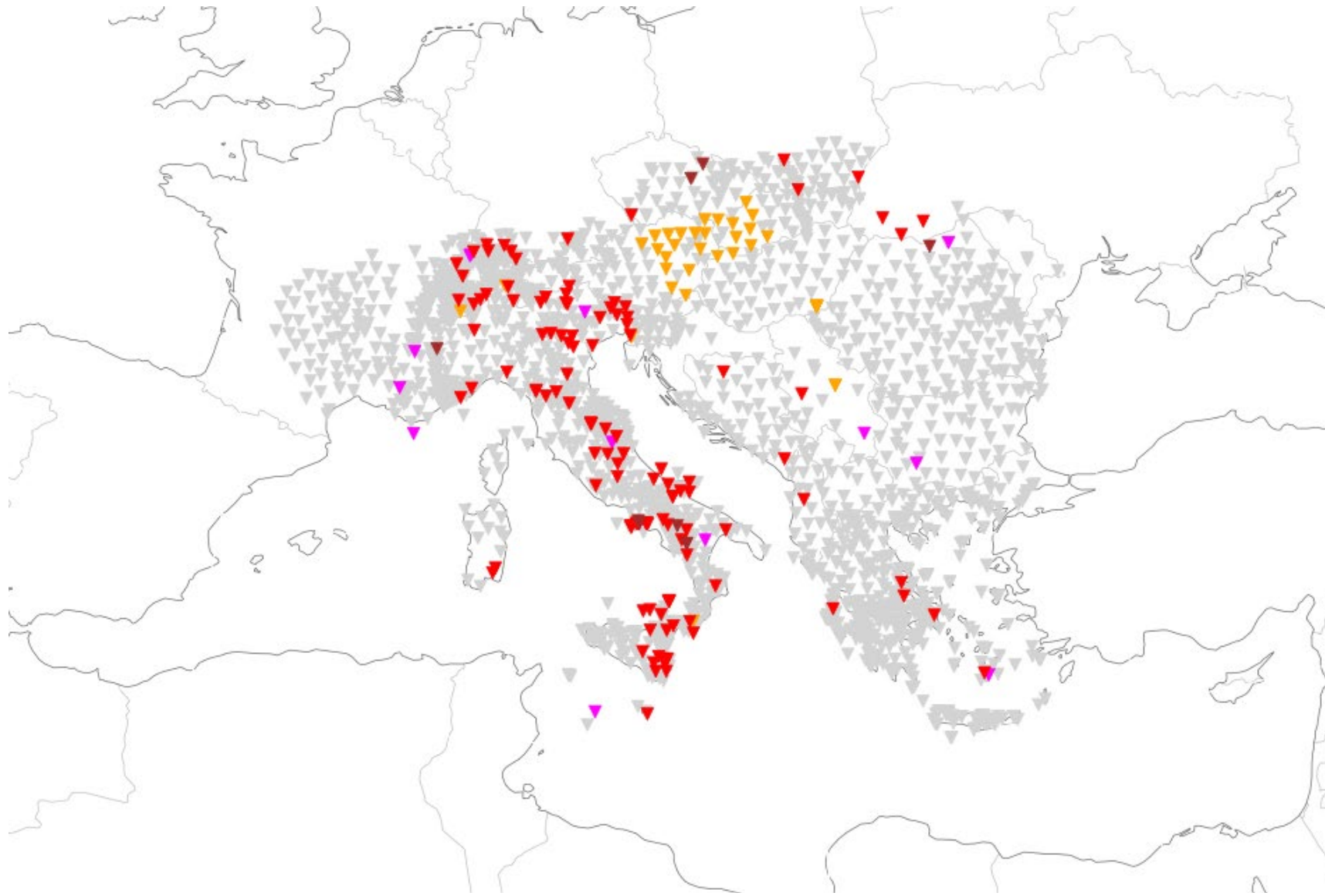
TESTS

- FDSN StationXML validation
- IRIS StationXML Validator - <https://github.com/iris-edu/StationXML-Validator/wiki>
 - validation levels: network, station, channel, response
- corner periods computed from the metadata responses
- checks for P&Z consistency
- cross-checks of corner periods from metadata and independent list

List of tested issues

	id	severity		
FDSN StationXML validation	2	-1	META_VALID	# metadata FDSNStationXML validation
	2.01	3	META_ATTRIB_MISS	# metadata - missing attribute
	2.02	1	META_CLOCK_VALUE	# metadata - wrong value format in ClockDrift
IRIS StationXML validator	3.111	3	META_IRIS_111	# metadata - Station epochs overlap
	3.112.1	1	META_IRIS_112.1	# metadata - Network and/or Station endDates set to future
	3.112.2	3	META_IRIS_112.2	# metadata - Station startDate before Network startDate
	3.212.2	3	META_IRIS_212.2	# metadata - Channel startDate before Station startDate
	3.222	4	META_IRIS_222	# metadata - Station and Channel Latitudes/Longitudes differ too much
	3.223	4	META_IRIS_223	# metadata - Station and Channel Elevations differ too much
	3.304	1	META_IRIS_304	# metadata - no Sensor Description in Channel
	3.310	3	META_IRIS_310	# metadata - Channel.startDate should exist and be before Channel.endDate
	3.320	4	META_IRIS_320	# metadata - missing Azimuth and/or Dip
	3.321	4	META_IRIS_321	# metadata - invalid InputUnits (first Stage) and/or OutputUnits (last Stage)
	3.332	1	META_IRIS_332	# metadata - mis-orientation >= 5, component code should be one of [123]
	3.402.1	0	META_IRIS_402.1	# metadata - case inconsistency in Input/OutputUnits
	3.402.2	3	META_IRIS_402.2	# metadata - invalid Input/OutputUnits
	3.402.3	4	META_IRIS_402.3	# metadata - invalid InputUnits in the first Stage
	3.403	4	META_IRIS_403	# metadata - Input/OutputUnits do not follow in Stages
	3.404	5	META_IRIS_404	# metadata - missing Decimation and/or StageGain in Stage(s)
	3.410	4	META_IRIS_410	# metadata - missing Value in InstrumentSensitivity
	3.411	3	META_IRIS_411	# metadata - InstrumentSensitivity.Frequency must be < SampleRate/2
	3.414	4	META_IRIS_414	# metadata - StageGain.Frequency is zero in undue Stage
	3.416	4	META_IRIS_416	# metadata - missing InstrumentSensitivity in Response
	3.420	3	META_IRIS_420	# metadata - missing at least one Decimation in Stages
	3.421	4	META_IRIS_421	# metadata - output samplerate from Stages not equal to Channel.SampleRate
	3.422	4	META_IRIS_422	# metadata - samplerates do not follow in Stages
Response checks, corner periods, band codes	4	5	META_RESP_FAIL	# metadata response failure
	4.01	5	META_NO_POLES	# metadata - no Poles in PolesZerosResponseStage
	4.02	5	META_LAST_STAGE_FAIL	# metadata - the last response stage is faulty
	4.03	3	META_REAL_POLE_POSITIVE	# metadata - real(Pole) >= 0 in the analog PolesZerosResponseStage
	4.04	5	META_POLES_NOT_CONJUG	# metadata - not conjugated Poles in the PolesZerosResponseStage
	4.05	3	META_BAND_SAMPL_RATE	# metadata - incorrect channel band due to sampling rate
	4.06	4	META_BAND_BB_SP	# metadata - channel band confusion in broadband and short-period
	4.07	2	META_CORNER_TABLE	# metadata - corner confusion between real and tabled values

IRIS StationXML validator without notifications

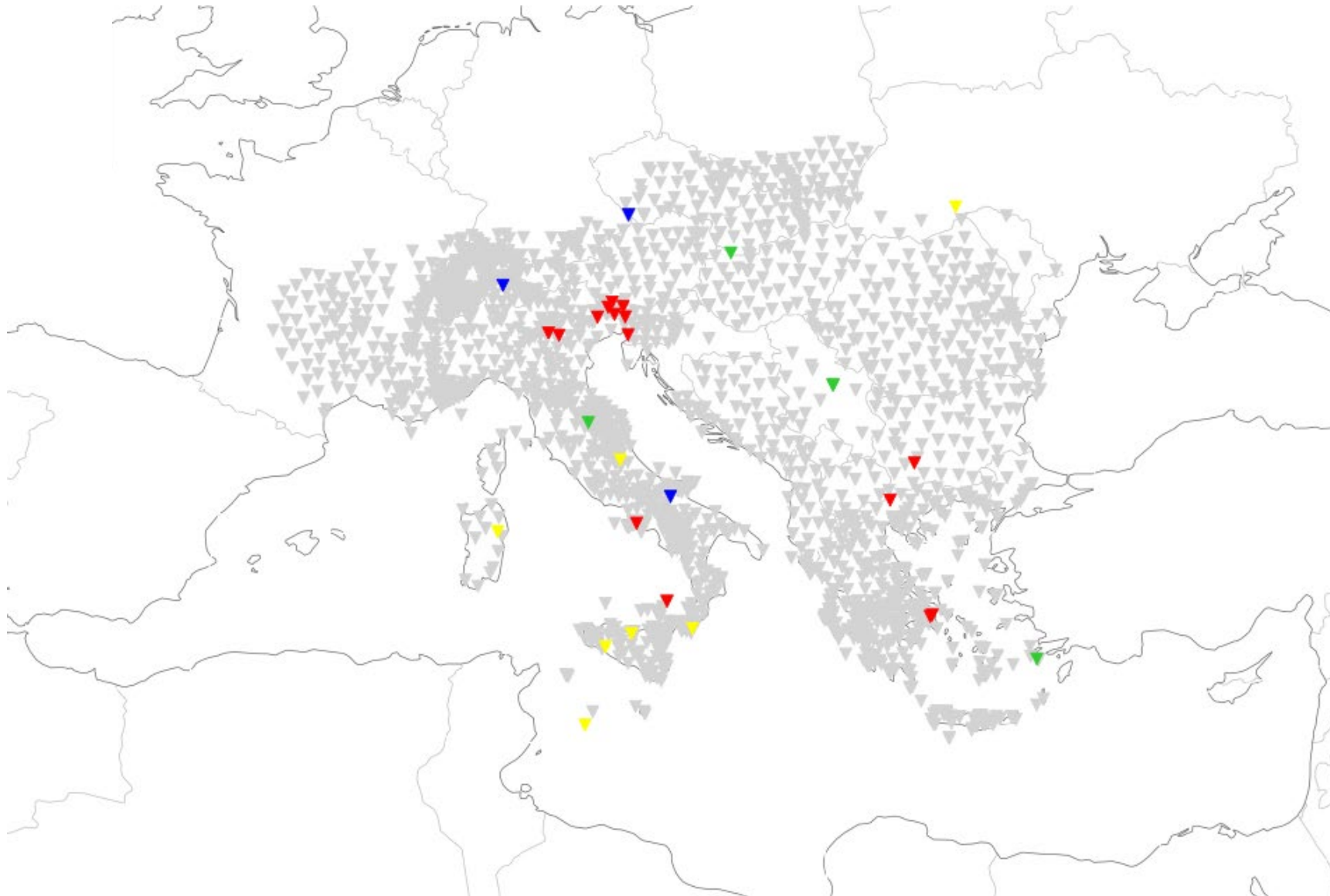


- network level
(1.4 %)
- station level
(0.6 %)
- channel level
(0.4 %)
- response level
(5.7 %)

total: 8 %

errors & warnings
only 2022+

Another response checks



- ▼ response failure
(15 stations)
- ▼ no Poles in the sensor
response stage
(3 stations)
- ▼ complex Poles not
conjugated
(4 stations)
- ▼ not negative real Pole
parts in the sensor
stages
(7 stations)

total: 1.6 %

errors
only 2022+

Corner period, band codes

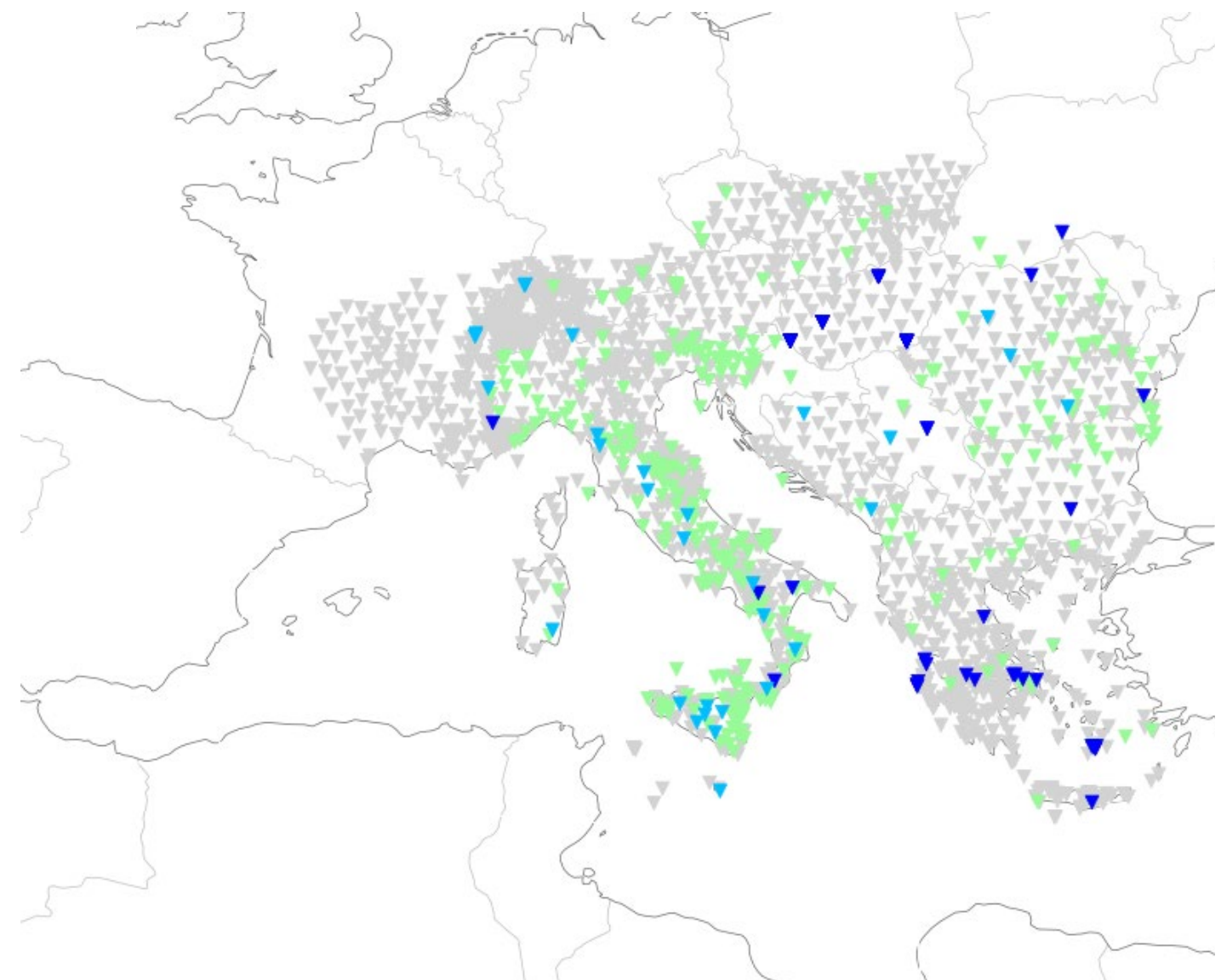
sampling
frequency
20 Hz

+

corner
period
60 s

=

band
code
BHZ



Band Code			
The first letter specifies the general sampling rate and the response band of the instrument. (The "A" code is reserved for administrative functions such as miscellaneous state of health.)			
Band code	Band type	Sample rate (Hz)	Corner period (sec)
F	...	≥ 1000 to < 5000	≥ 10 sec
G	...	≥ 1000 to < 5000	< 10 sec
D	...	≥ 250 to < 1000	< 10 sec
C	...	≥ 250 to < 1000	≥ 10 sec
E	Extremely Short Period	≥ 80 to < 250	< 10 sec
S	Short Period	≥ 10 to < 80	< 10 sec
H	High Broad Band	≥ 80 to < 250	≥ 10 sec
B	Broad Band	≥ 10 to < 80	≥ 10 sec
M	Mid Period	> 1 to < 10	
L	Long Period	≈ 1	
V	Very Long Period	≈ 0.1	
U	Ultra Long Period	≈ 0.01	
R	Extremely Long Period	≥ 0.0001 to < 0.001	
P	On the order of 0.1 to 1 day ¹	≥ 0.00001 to < 0.0001	
T	On the order of 1 to 10 days ¹	≥ 0.000001 to < 0.00001	
Q	Greater than 10 days ¹	< 0.000001	
A	Administrative Instrument Channel	variable	NA
O	Opaque Instrument Channel	variable	NA

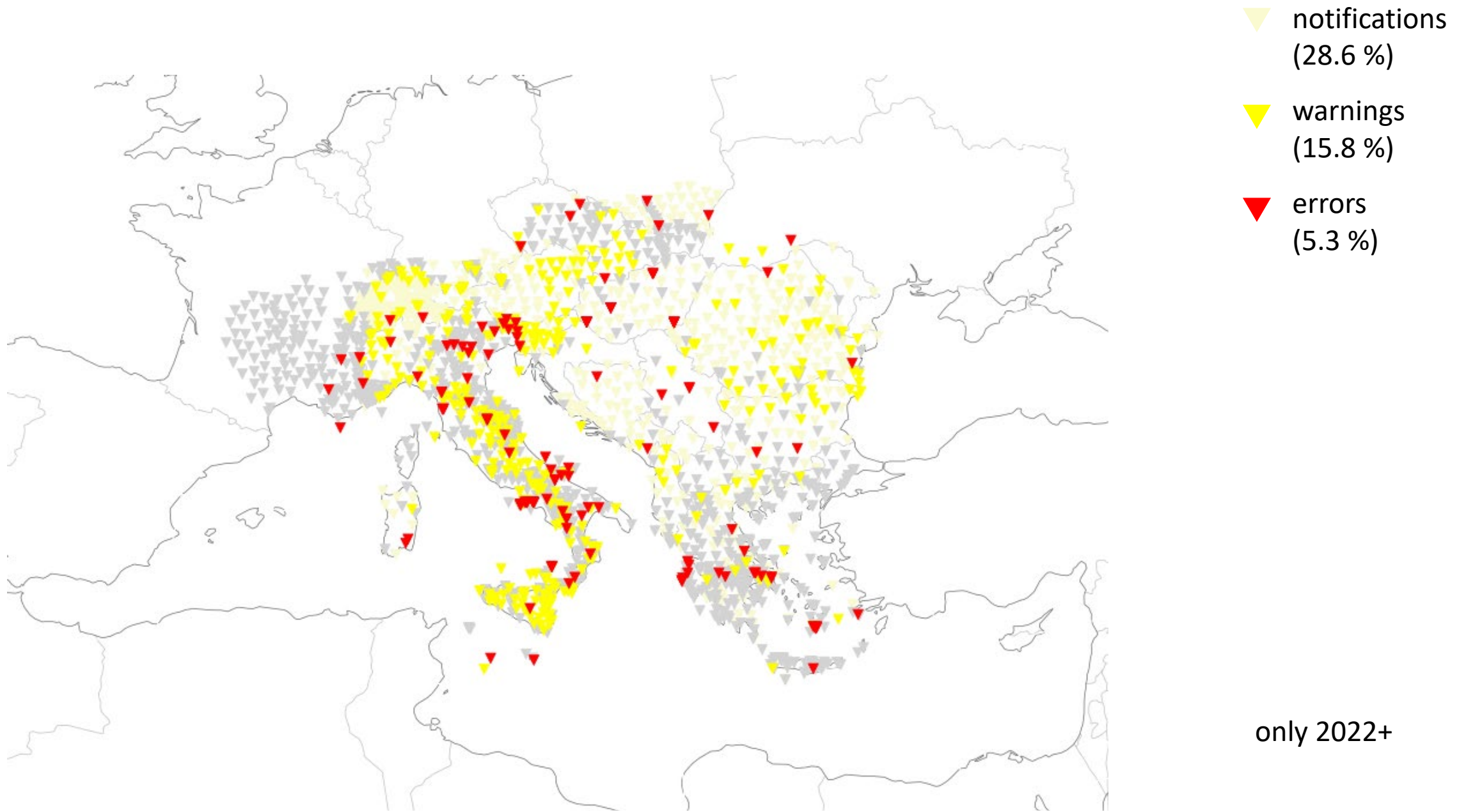
1. These are approximate values. The sample rate should be used for the correct Band Code.

[SEED manual](#)

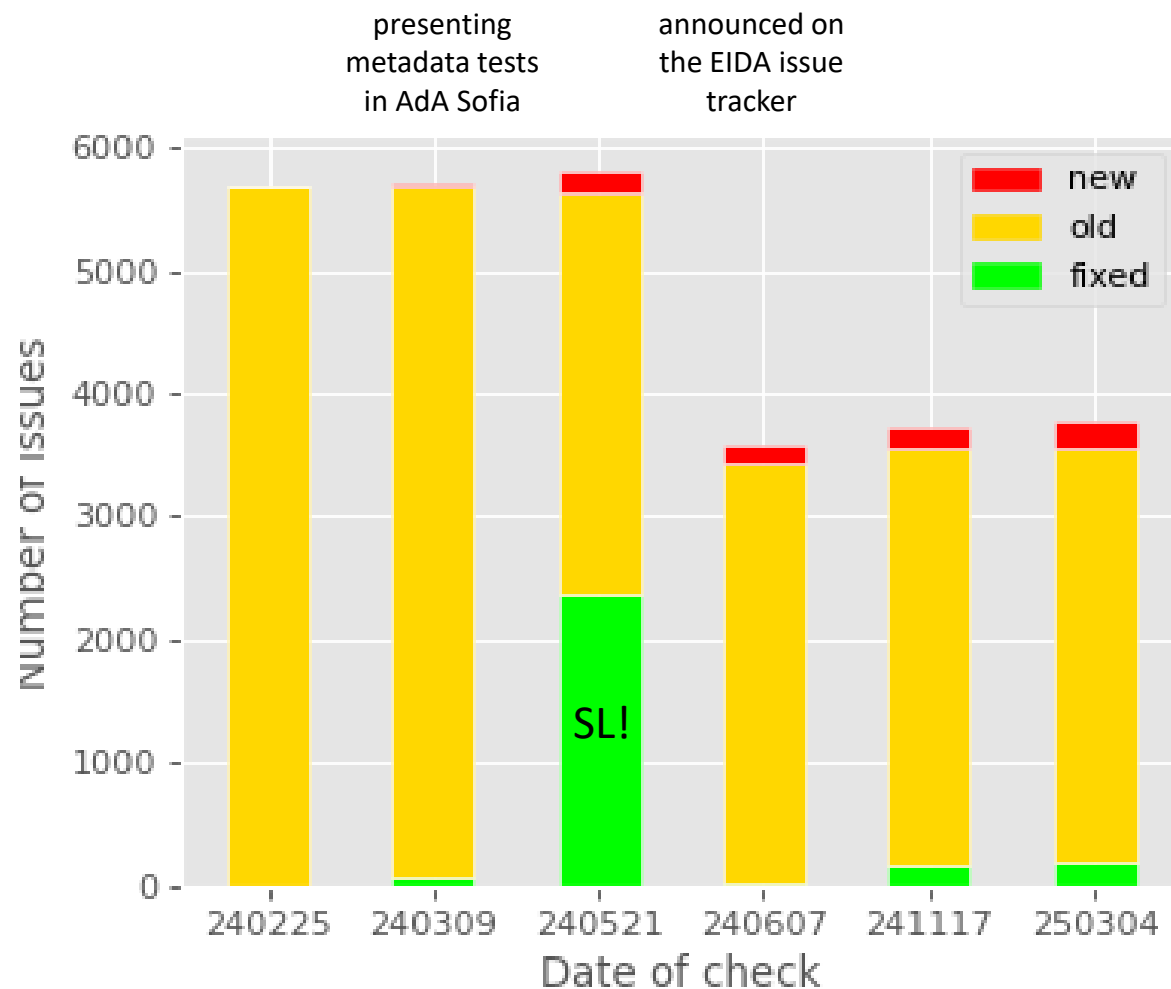
- different corner periods in metadata and the list (17.5 %)
- wrong band code for a given sample rate (1.7 %)
- wrong band code for broad-band/short-period (1.7 %)

total: 21 %

Issue severity levels



Trends in the metadata tests



- first results of the metadata tests presented in the AdA workshop in Sofia 2024
- then big metadata corrections in the SL network
- announcement at the EIDA issue tracker on Jun 3, 2024
<https://github.com/EIDA/userfeedback/issues/166>

What next?

<https://orfeus-eu.org/data/eida/webservices/wfcatalog/>

Incorporate EIDAWS WFCatalog!

- detailed information on the contents of waveform data (mseeds) in EIDA
- easy and fast download from EIDA
- daily metrics, such as:

num_gaps, num_overlaps, sum_gaps, sum_overlaps, max_gap, max_overlap

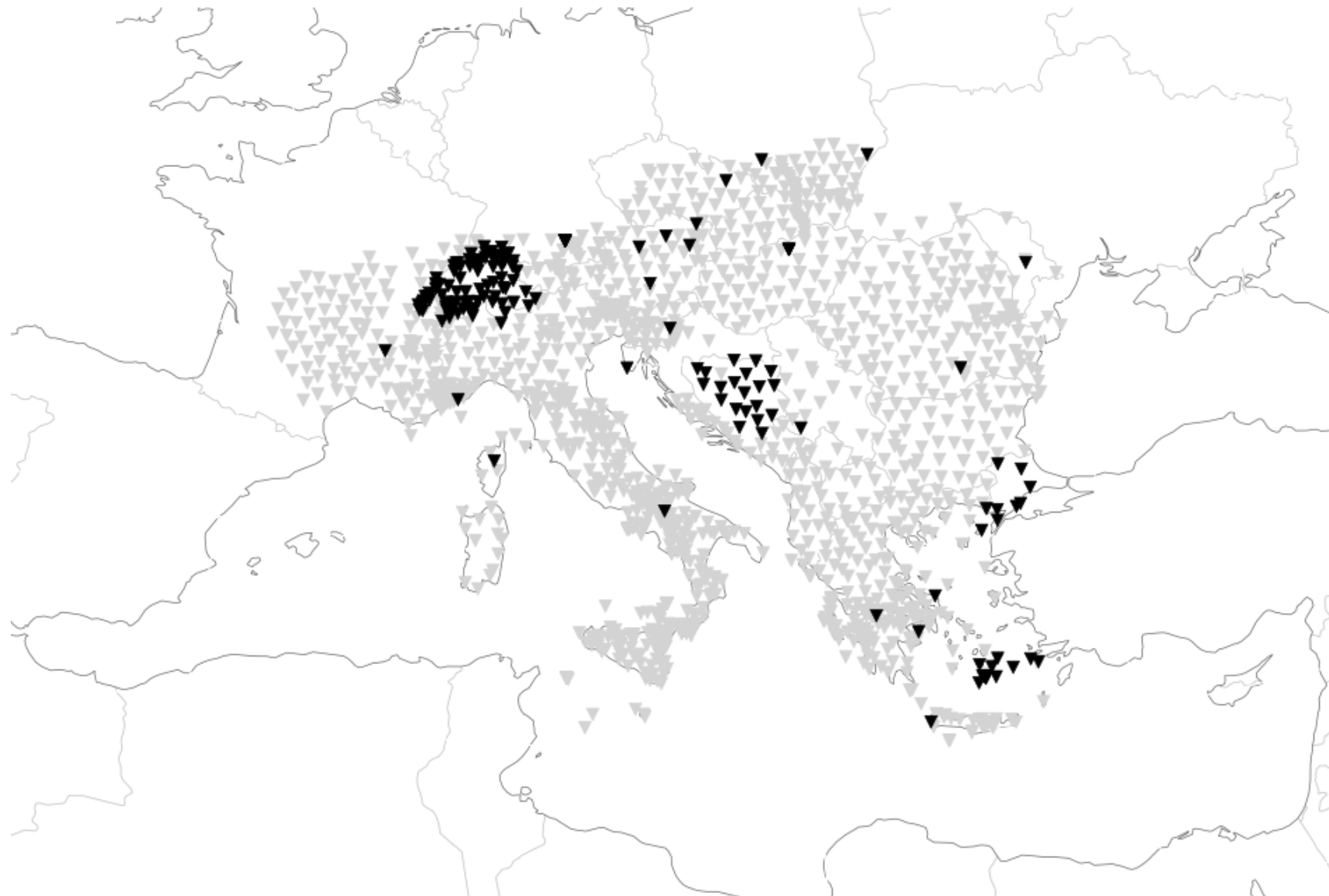
record_length, sample_rate, num_records, percent availability

encoding, format, quality

sample_min, sample_max, sample_mean, sample median, sample stdev, sample_rms,

sample_lower_quartile, sample_upper_quartile

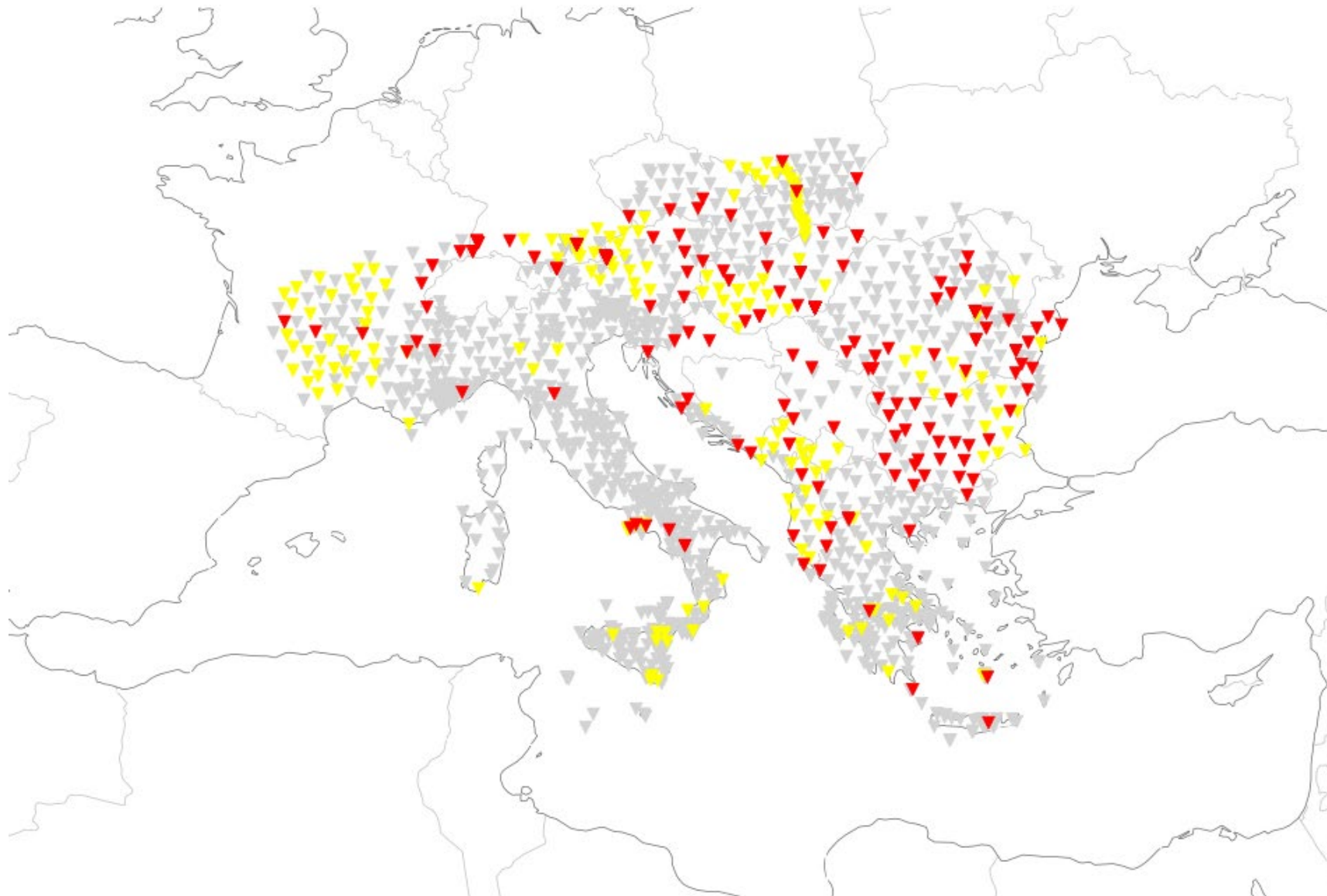
WFCatalog availability



▼ no catalog
obtained
(9.9 %)

stations with a BB
channel during 2022-
2025

Fitting data and metadata

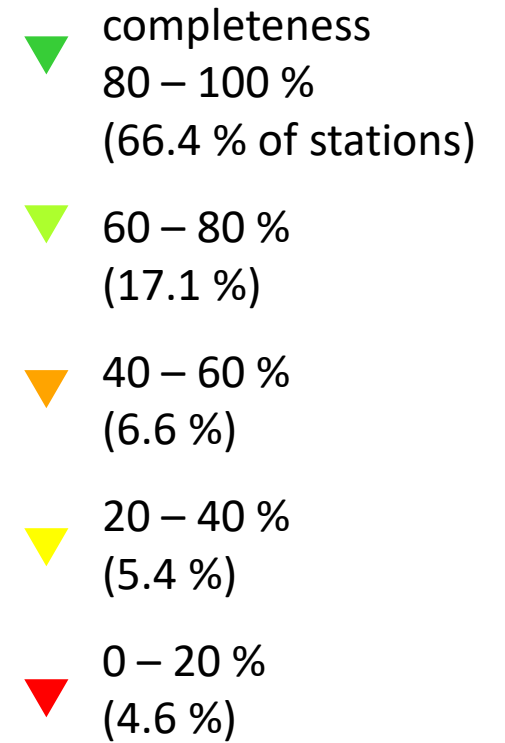
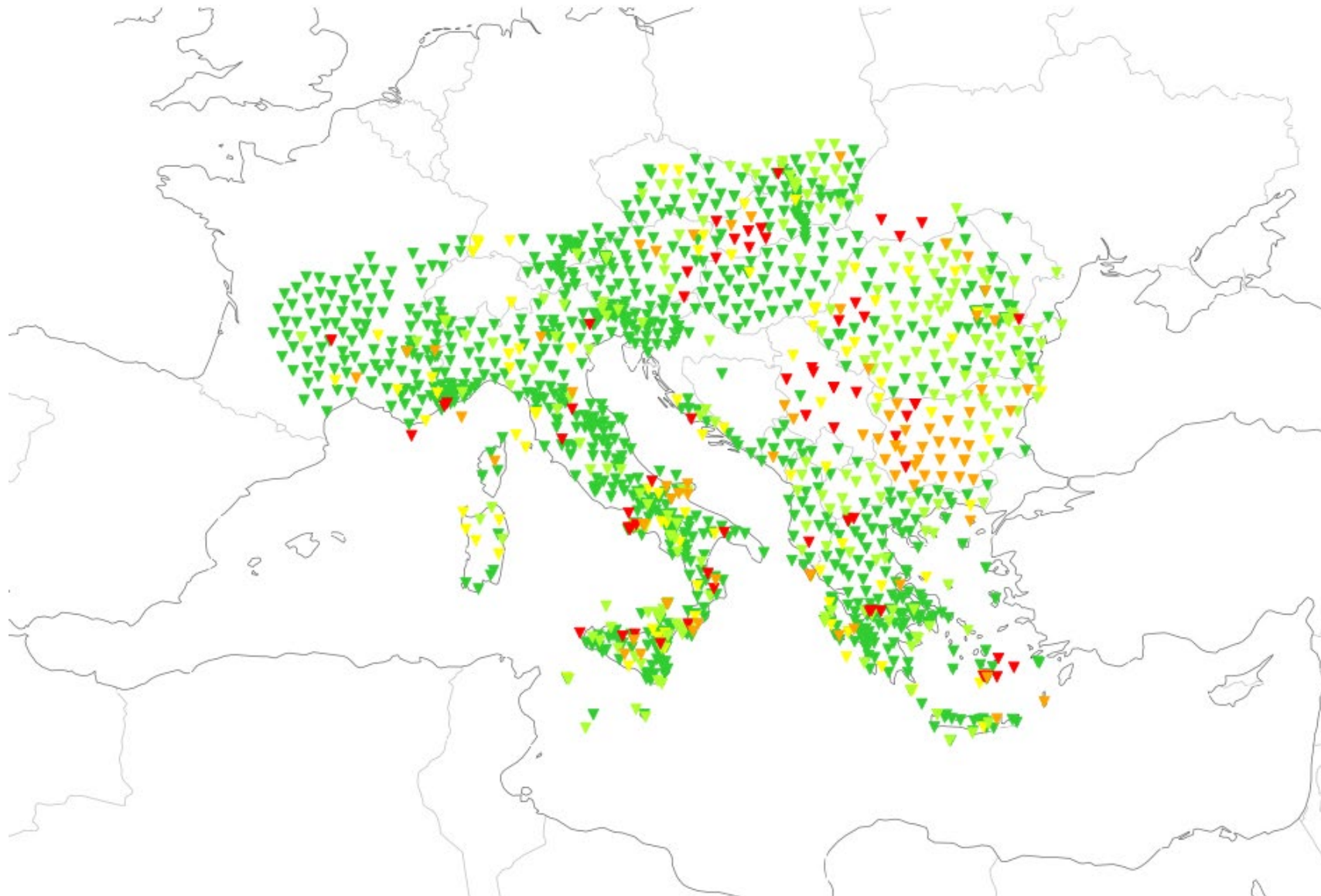


Mseed BB channel
not included in
metadata

- ▼ for less
than 6 days
(11 %)
- ▼ longer
(13 %)

WFCatalog versus
StationXML

Data completeness (via WFCatalog)



WFCatalog is not up to date!
delay 3-5 months at
some stations

Saturated signal offset (via daily medians)

Tested period:
2022-01-01 – 2023-03-21
445 days

Saturation condition:
 $\text{abs}(\text{median}) > 1\text{e}6$

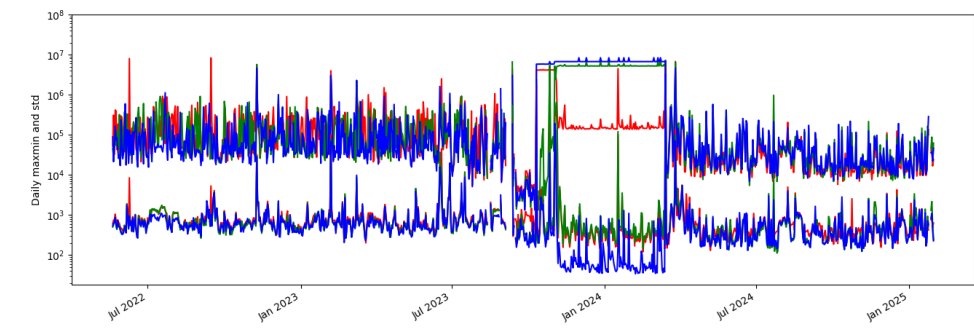
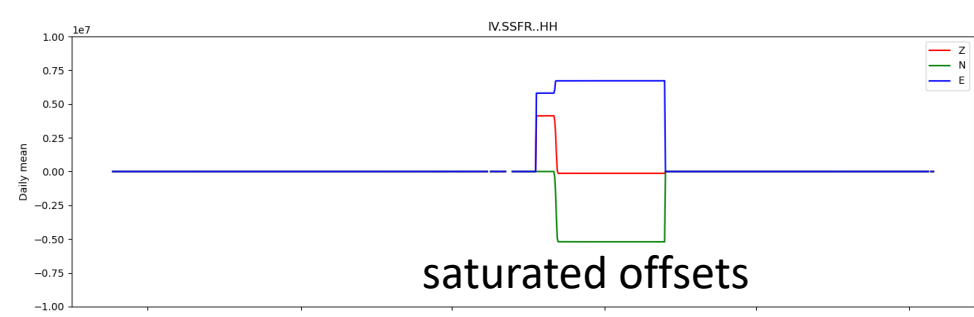
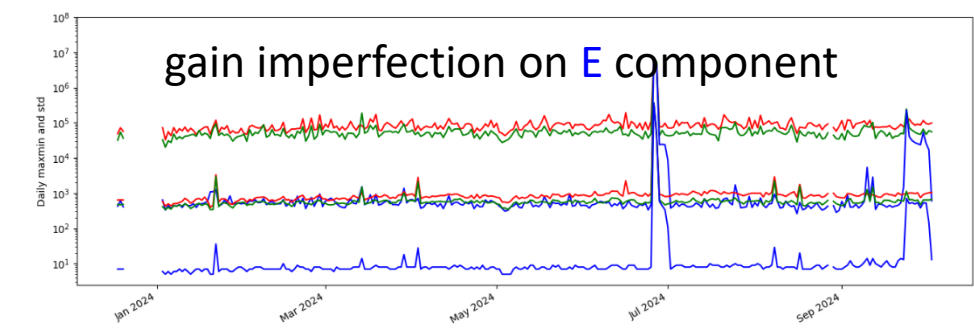
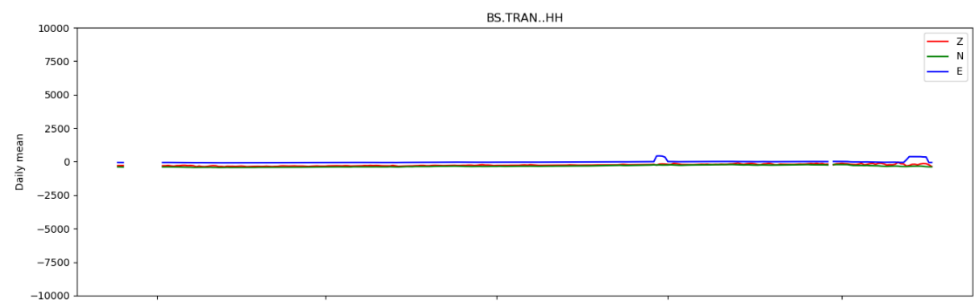
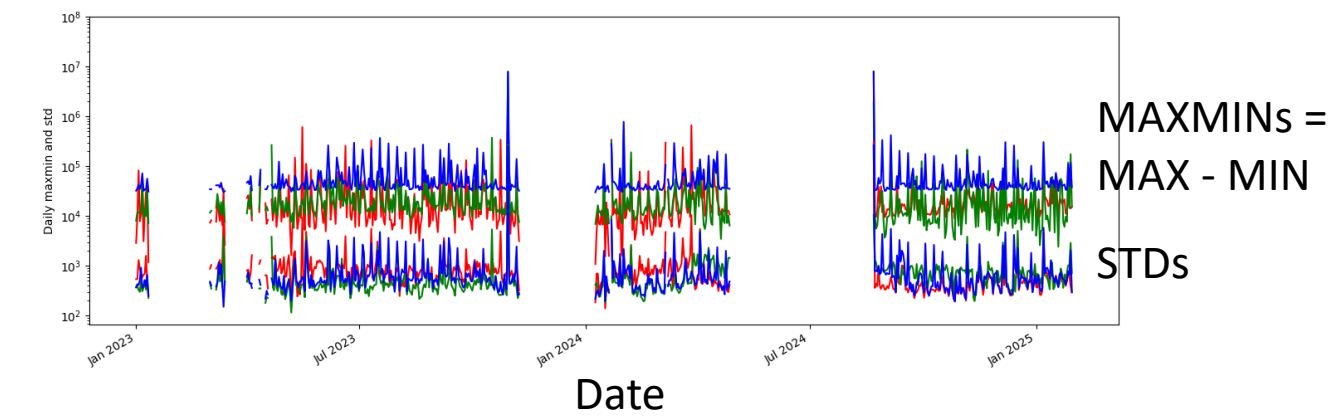
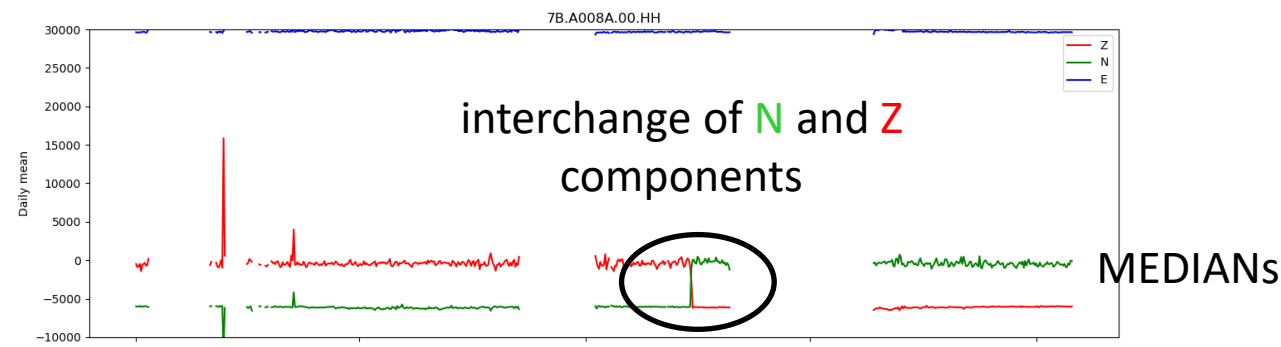


- ▲ all stations (861)
- ▲ 10+ days: 79/861 9 %
- ▲ 100+ days: 43/861 5 %
- ▲ 300+ days: 28/861 3 %

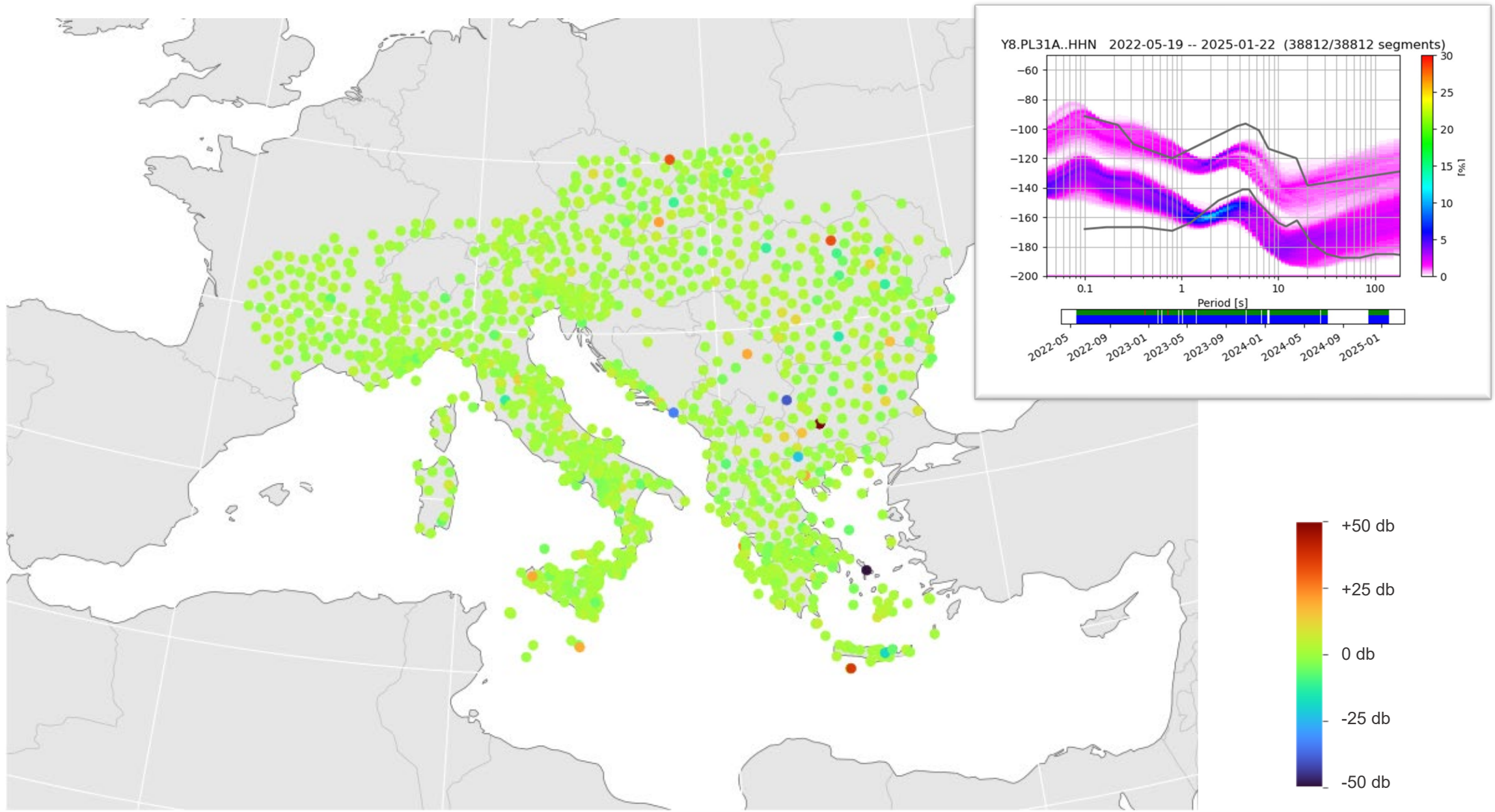
What is signal maximum?

24 bit ADC = $8.4\text{e}6$
32 bit ADC = $2\text{e}9$

Metrics over time



WFC – amplitude gains (ratios of E-N daily maxmins calibrated by sensor sensitivities)



Maximal amplitudes from a seismic event

Amplitude gains

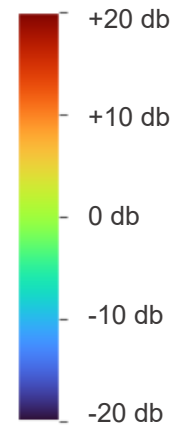
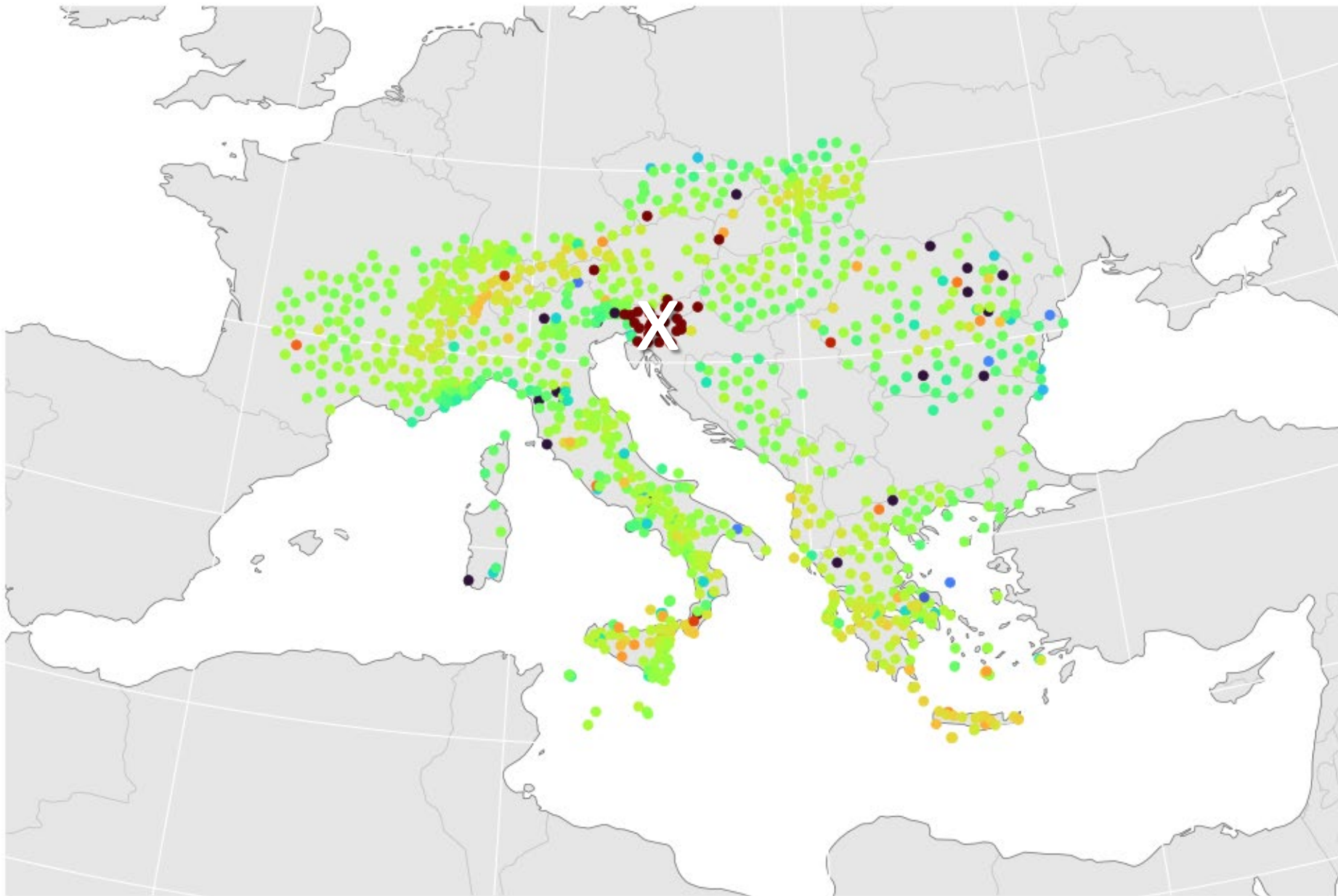
Morocco 6.8

2023-09-08T22:11

N31.055°, W8.389°

depth 19.0 km

Z component
filtered 20-60 s



Concluding remarks



- We can identify issues in station metadata , results are published:
<https://github.com/PetrColinSky/AdriaArray>
- Thank to EIDA for WFCatalog, other service needed: PPSD!
- Time to manage other issues, especially in data: saturated signal or signal clipping, sensor mis-orientations, amplitude misgains, ... (Kolínský et al., 2025b)
-> action for ~~ORFEUS UAG~~ AdriaArray - WG 3 'Data QC' ?
- There are many methods for identifying problems in data and metadata. Now, we need procedures for presenting the results from these methods, sharing them, and determining what can be fixed (and pushing those responsible to fix it).

Issue severity levels

NOTIFICATIONS

0

1

not important at all

WARNINGS

2

3

station operators could
check their metadata

ERRORS

4

5

station operators **should**
check their metadata

- case inconsistency in Input/OutputUnits

- wrong value format in ClockDrift
- Network and/or Station endDates set to future
- no Sensor Description in Channel
- **mis-orientation** ≥ 5 , **component code should be one of [123]**

?

- confusion between real and tabled values of corner periods

- Station epochs overlap
- Station startDate before Network startDate
- invalid Input/OutputUnits
- missing at least one Decimation in Stages
- incorrect channel band due to sampling rate

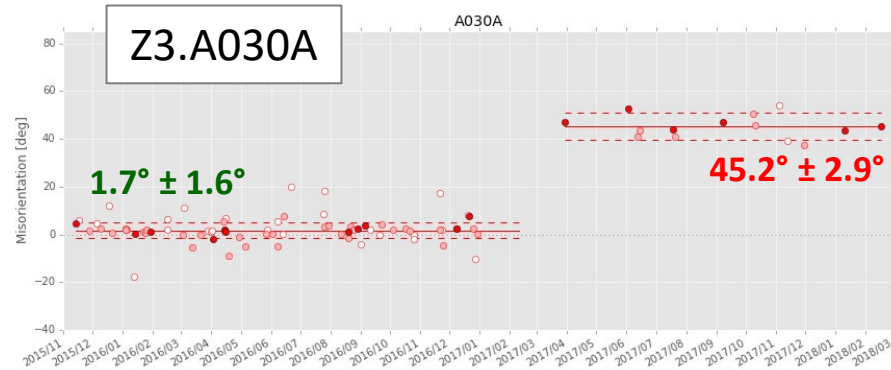
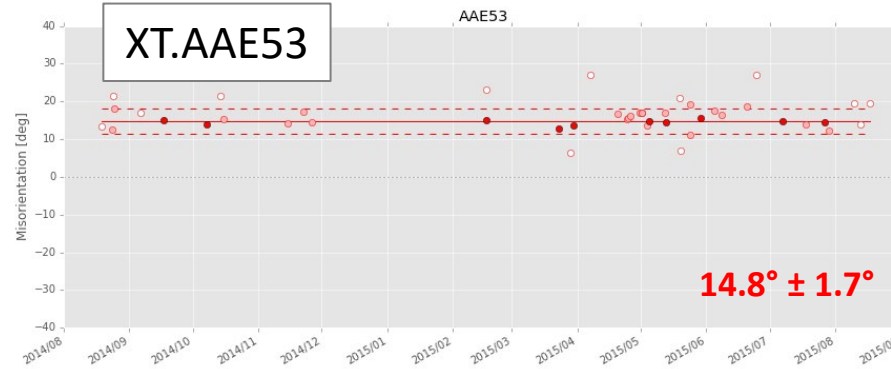
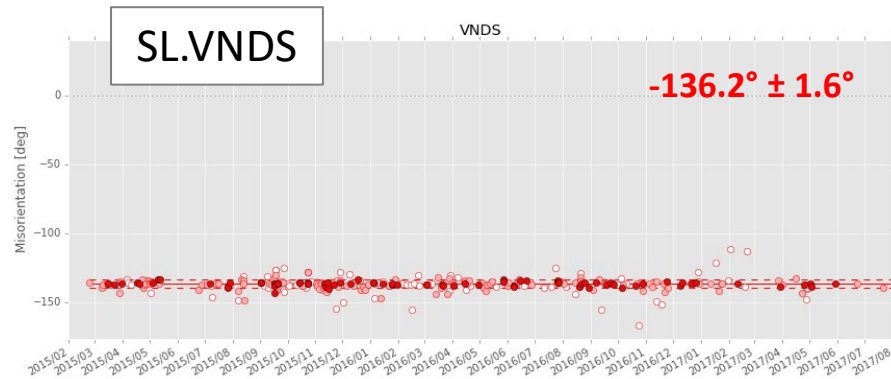
- Station and Channel Latitudes/Longitudes differ too much
- missing Azimuth and/or Dip
- missing Value in InstrumentSensitivity
- missing InstrumentSensitivity in Response
- channel band confusion in broadband and short-period

- missing Decimation and/or StageGain in Stage(s)
- metadata response failure
- no Poles in PolesZerosResponseStage
- the last response stage is faulty
- not conjugated Poles in the PolesZerosResponseStage

Lvl 0 is not shown
in our tests!

Sensor mis-orientations

Sensor orientation



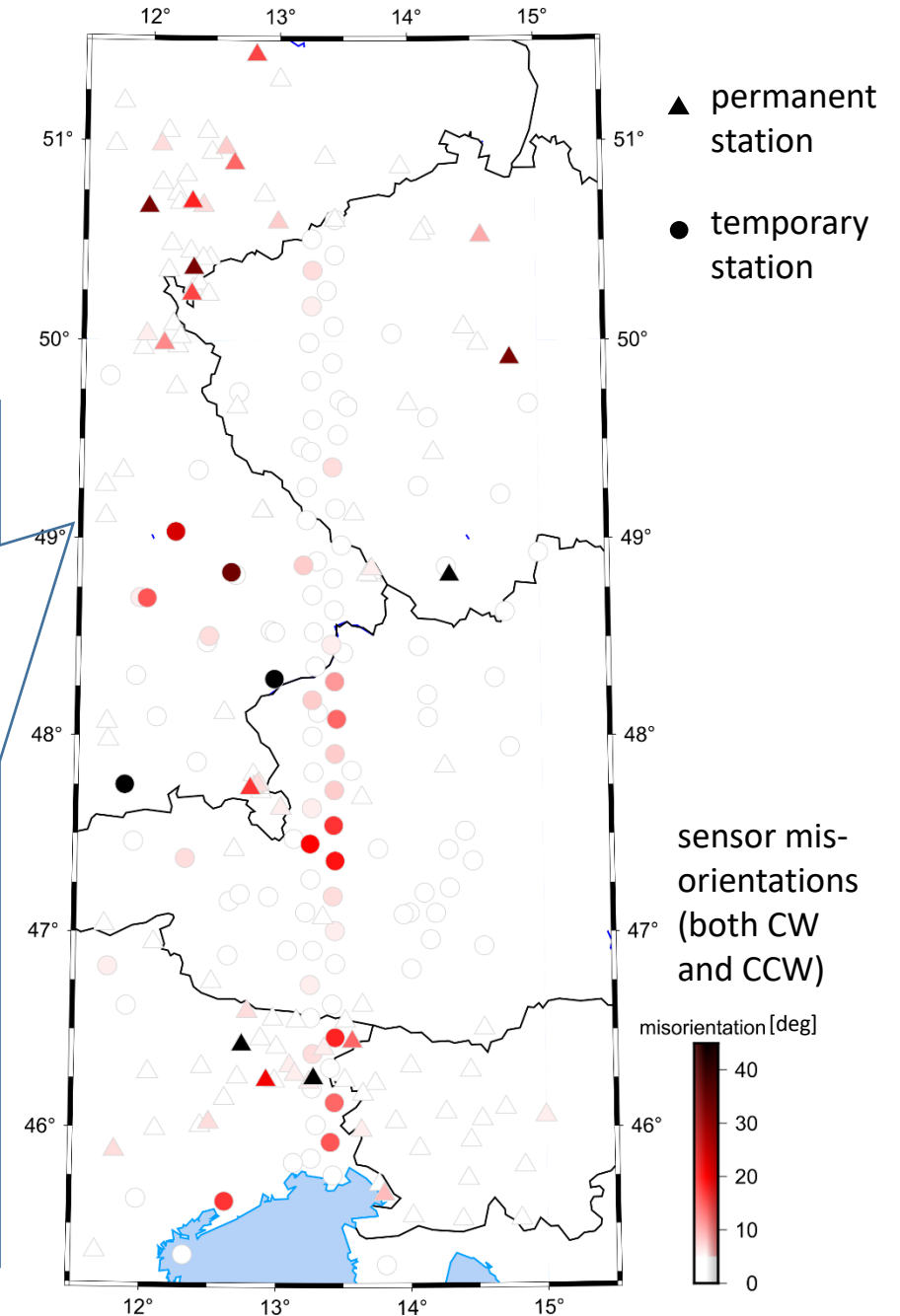
2014-2018: Rayleigh wave method (multi station)

2014-2023: SKS polarization method (single station)

Deviations of sensor orientations from the official orientations given in metadata (from EIDA, 2023-06-05).

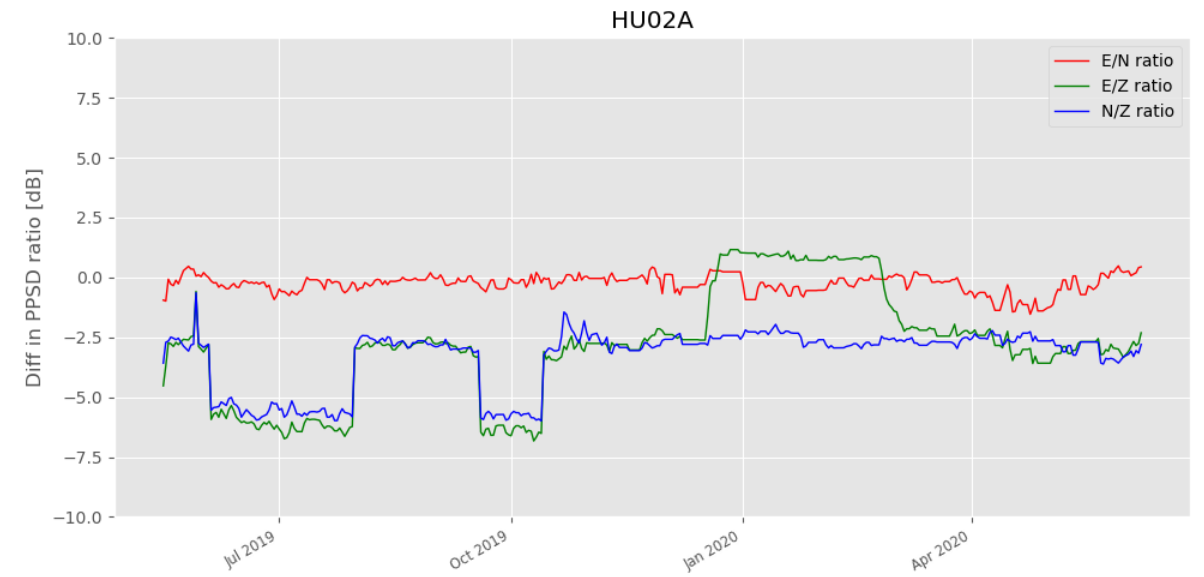
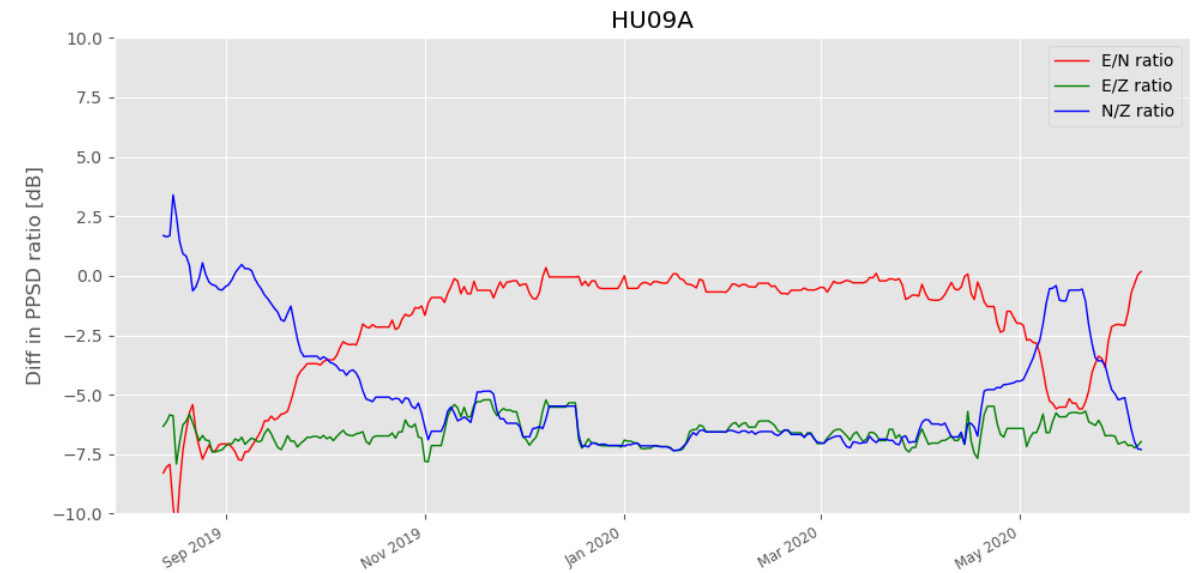
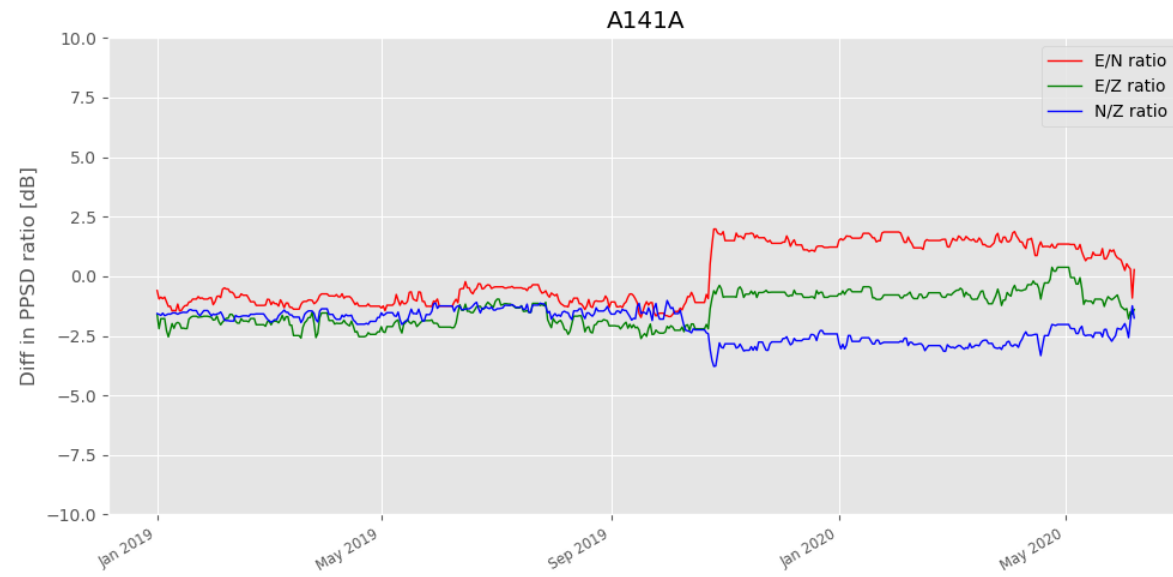
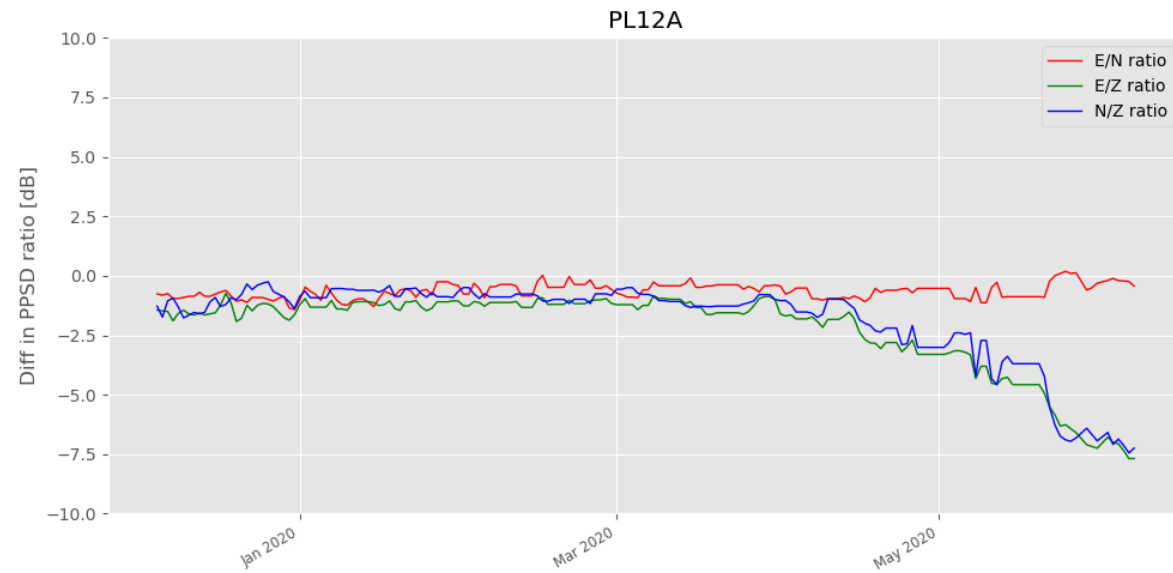
Number of stations with sensor mis-orientations:

$\geq 5^\circ$	77/250	31 %
$\geq 10^\circ$	29/250	12 %
$\geq 20^\circ$	12/250	5 %
$\geq 30^\circ$	9/250	4 %
$\geq 45^\circ$	5/250	2 %
$\geq 90^\circ$	3/250	1 %



PACASE - amplitude gain issues

- temporal variations of PPSD ratios (for $T = 5s$)
- global trend subtracted



Timing issues - examples

